Reading Without Looking: Position-Resolved Evidence of Visual Bypass in a Small Document OCR Model

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Abstract

Confidence scores from document OCR models are used to decide which predictions a human should check. We audit that signal in a small ($\approx$19.5M parameter) encoder–decoder OCR model trained from scratch with no prior exposure to any Indic script, across three independently trained seeds. We first establish what the model can do: on held-out real Hindi scans its grapheme character error rate is 0.985, and on blank white images it is 0.949. The model does not read real document images, and the real–blank difference does not survive a seed-clustered paired test ($t = 0.257$, $p = 0.822$). We then localise the failure. Under teacher forcing the ground-truth token is never the model's argmax at generation position 0 (0 of 180 sequences), and the geometric mean probability assigned to the correct first symbol is 2.2×10^{-11} on real images and 6.3×10^{-11} on blank, roughly eight orders of magnitude below the uniform value of 2.7×10^{-3} over a 367-symbol vocabulary, while self-generated max-softmax confidence at that same position is approximately 0.90. From position 2 onward the model recovers to a mean log-likelihood between -0.15 and -0.48, above a trigram grapheme language model baseline of -0.99, and does so identically whether the image contains text or nothing. Deleting the encoder changes the output distribution substantially (mean KL 1.075 nats, argmax flips at 8.2% of decoding steps) but leaves its peak near unity: 96.7% of the divergence sits at those flipped positions while agreeing positions retain a mean confidence of 0.998. Finally, the confidence signal is not globally broken. On synthetic in-distribution lines, where the model attains 18.3% line accuracy, confidence discriminates correct from incorrect transcriptions at AUROC 0.838 while remaining badly miscalibrated at a mean of 0.994; on real scans, where accuracy is zero, it carries no usable correctness signal. We read this as a model whose certainty tracks the linguistic plausibility of what it generates rather than the state of its own perception, and we report the position-resolved likelihood profile as a cheap diagnostic for the failure mode.

1 Introduction

Document OCR is often the first automated step in processing multilingual records, and production pipelines route low-confidence predictions to human review. The design assumes that when a model is unsure, it says so.

That assumption is unreliable for large vision–language models. Recent work establishes *visual ungroundedness*: models produce fluent, confident, sometimes correct output driven entirely by language priors, with the image contributing nothing, and existing confidence estimators cannot detect this [Khanmohammadi et al., 2026]. Related work describes the same hazard as perception bypass or visual shortcut [Xu, 2026], and multilingual OCR benchmarks report that failure on unfamiliar scripts is not silent: models hallucinate fluent text in familiar scripts rather than abstaining [Kargaran et al., 2026].

We take that as established and ask a narrower question that autoregressive transcription makes answerable: **where in the generated sequence does visual dependence live, and can its absence be localised without training a probe?**

The question has a structure here that short-answer visual question answering does not offer. When a decoder transcribes a line token by token, every position after the first can draw on two sources: the image, through cross-attention, and the preceding tokens, through the language prior over the training script. These are confounded. Position 0 is not. At position 0 there is no prefix, so any probability placed on the correct first symbol must have come from the image. Position 0 is a single, theoretically motivated test point for visual dependence requiring no auxiliary labels beyond the ground-truth text, no trained detector and no comparison model.

We build a small encoder–decoder OCR model from scratch rather than fine-tuning an existing checkpoint, so that "this model has never seen this script" is a claim we can make rather than assume. We then run four interventions on fixed checkpoints across three seeds: encoder output deletion, blank and unfamiliar-script input, teacher-forced ground-truth likelihood, and a positional decomposition of that likelihood.

Contributions. (1) We establish by direct measurement that the instrument does not read real document images (grapheme CER 0.985 real against 0.949 blank, a difference that does not survive seed clustering), and treat this as a finding to be explained rather than a caveat to be buried; every subsequent claim is scoped by it. (2) We give a position-resolved grounding diagnostic for autoregressive transcription, motivated by the unconfounded status of position 0 and requiring only ground-truth text and two forward passes. (3) We decompose encoder ablation into distributional *shape* (KL divergence, large, 96–99% concentrated at argmax flips) and *peak height* (max-softmax, 0.998 at agreeing positions), showing that the quantity deployed as confidence is the one component of the output distribution that visual ablation leaves near ceiling. (4) We identify a regime in which the same confidence signal does discriminate correctness (AUROC 0.838 at 18.3% accuracy) while remaining uncalibrated, giving a within-instrument contrast rather than a purely negative result. (5) We show that 16.9–20.4% of non-exact predictions from off-the-shelf OCR engines on real Devanagari scans are Unicode encoding variants rather than misreadings, a precondition for any grounding claim measured against exact-match references.

What this paper does not claim. We did not discover that confidence fails to track visual grounding; that is prior work [Khanmohammadi et al., 2026]. We make no claim about production-scale OCR systems. And we do not yet have a positive control, a model with demonstrated reading ability on which the same diagnostic returns a different answer; Section 10 states what that costs.

2 Related Work

Visual bypass and language priors. That a multimodal model can answer without using the image originates in visual question answering and has been rediscovered repeatedly. Contemporary work reports blind-image ablations in which replacing informative frames with blank ones leaves accuracy unchanged or improved, and treats this as a documented hazard rather than a novel observation [Xu, 2026]. A blind language model ignoring image evidence has been shown to outperform prior art on some image–text retrieval benchmarks [Lin et al., 2024].

Confidence estimation under ungroundedness. BICR [Khanmohammadi et al., 2026] is the closest prior work. It establishes that existing confidence estimators cannot separate grounded from ungrounded predictions, and proposes a probe trained on real-image hidden states, regularised by a ranking loss against a blacked-out view, evaluated across five large vision–language models. Our contribution differs in being positional rather than learned, training-free, and aimed at sequence transcription rather than short-answer generation. We do not offer a better estimator; we offer a cheaper diagnosis of a specific structural cause.

Abstention and probing in OCR. Latent-state and attention probes have been trained for OCR-error abstention, including visual-focused probes motivated by the informativeness of attention to visual tokens [Yao et al., 2025]. HALP predicts hallucination risk before generation from internal representations, reaching up to 0.93 AUROC on several models [Kogilathota et al., 2026]. Both require supervision; the diagnostic here does not.

Multilingual OCR and grapheme-aware evaluation. GlotOCR Bench finds that models hallucinate fluent text in familiar scripts rather than abstaining, with even the best model transcribing under 8% of low-resource-tier sentences below 5% CER [Kargaran et al., 2026]. A Devanagari stress-test benchmark finds that synthetic renders badly overstate quality, that nine of ten evaluated systems collapse on real scans, and that English OCR strength does not predict Indic performance [Singh, 2026]. `grapheme-kit` extends evaluation metrics from Unicode code points to grapheme clusters and shows via an OCR case study that grapheme-level metrics evaluate complex scripts more faithfully [Nisfer et al., 2026]; Section 4 applies that argument rather than claiming it.

3 The Instrument

A small encoder–decoder vision–language model, approximately 19.5M parameters. A Vision Transformer encoder trained from random initialisation, with no pretrained weights, produces patch-level visual features. An autoregressive decoder cross-attends to those features and emits one grapheme cluster per step. The output vocabulary is defined over grapheme clusters, the visually atomic units of the script, rather than byte-pair-encoded subwords, so one output token corresponds to one decision about what was read. Vocabulary size is $|V| \approx 367$, fixing the uniform per-token probability at 2.725×10^{-3} and making the position-0 results in Section 8 interpretable against chance. Three seeds were trained independently under identical hyperparameters, a design adopted after a single-seed result failed to replicate (Appendix A).

The training pipeline includes a renderer with a glyph-frequency dial supporting three target distributions (natural, flattened, inverted), achieving total-variation distance ≤ 0.08 from target in all three. As a measurement and control instrument it worked; as a data generator it did not. Line accuracy across the three conditions is 18.3% $\pm$ 5.1 (natural), 0.33% $\pm$ 0.6 (flattened) and 0.67% $\pm$ 0.6 (inverted), seed means over three seeds. With the dependent variable at the floor in two of three conditions, a fixed-effects decomposition of exposure against intrinsic script complexity is uninterpretable, and we withhold it (Section 10).

4 What Counts as an Error

Indic orthography admits multiple valid representations of visually identical text, so exact-match scoring conflates encoding variation with misreading. Every claim below is measured against ground-truth strings, so the reference must be established first. We distinguish three categories. **Tier 1 (encoding)**: deterministic byte-level variation with no difference in meaning or rendering, covering Unicode normalisation forms, zero-width joiner and non-joiner variants around conjuncts, danda versus period, script-native versus Latin digits, and anusvara versus homorganic nasal under a fixed sandhi rule; computed by a deterministic normalisation function with no judgment involved. **Tier 2 (phonetic)**: whether two spellings represent the same intended word, assessed by canonicalisation through ISO 15919 rather than a hand-enumerated rule list, validated against 38 hand-constructed pairs (23 positive, 15 negative or boundary). **Human residual**: genuine misreading, dropped diacritics, reading-order breaks, hallucinated or repeated content, assessed by grapheme-cluster alignment rather than code-point alignment.

Table 1: Tier 1 encoding variants among predictions that do not exactly match the reference, on real scanned Devanagari. The final column is the quantity of interest. PaddleOCR is shown for completeness only; $n = 10$ cannot support an engine-level claim.

Engine	n	Exact	Tier 1	Tier 1 among non-exact
Tesseract	180	15.6%	17.2%	**20.4%**
Surya	222	46.8%	9.0%	**16.9%**
PaddleOCR	10	0%	10.0%	10.0%

Between roughly one sixth and one fifth of what naive exact-match scoring calls an OCR error on these documents is an encoding variant, not a misreading. Two caveats belong with that number. First, a large share of non-exact predictions remains unreviewed (60.0% for Tesseract, 44.1% for Surya, 90.0% for PaddleOCR), so the Tier 1 shares are lower bounds on the encoding-artifact rate rather than final estimates. Second, Tier 2 resolves *zero* additional cases on all three engines: once Tier 1 normalisation is applied, phonetic spelling variation contributes nothing measurable on this corpus. We report the Tier 2 machinery because it was validated and applied, and its null result is itself informative about which kinds of orthographic ambiguity actually arise in Devanagari OCR output. All model-side results below use Tier-1-normalised references and grapheme-cluster alignment.

5 The Instrument Does Not Read Real Scans

We report this before any confidence result, because it scopes everything that follows.

Table 2: Grapheme CER by condition and seed, after Tier-1 normalisation with grapheme-cluster segmentation and Levenshtein distance. Lower is better; 1.0 corresponds approximately to a prediction sharing nothing with the reference.

Condition	Seed 0	Seed 1	Seed 2	Pooled
Hindi, real scans ($n = 60$/seed)	1.009	0.984	0.963	**0.985**
Blank white ($n = 60$/seed)	0.830	0.896	1.122	**0.949**
Ol Chiki, unseen ($n = 100$/seed)	0.974	0.906	1.029	0.970
Perso-Arabic, unseen ($n = 20$/seed)	0.958	0.906	1.110	0.991

An independent probe on a separate real-image set reproduces the ordering: 0.986 on real plain images against 0.942 on blank, $n = 180$ each. Three things follow.

First, **the model does not read real document images**. A CER near 1.0 is not partial reading; it is a transcription sharing almost nothing with the reference.

Second, **the nominal blank advantage is not robust and we do not build on it**. The sign is not stable across seeds: blank CER is lower at seeds 0 and 1 ($+0.180$ and $+0.088$) and higher at seed 2 (-0.159). A Wilcoxon signed-rank test on the 60 paired images is significant unclustered ($p = 0.025$) but not once clustered by seed ($t = 0.257$, $p = 0.822$). Generated length explains 23–24% of CER variance in both conditions, and on exact length-matched pairs ($n = 54$) the gap is $+0.039$ with $p = 0.197$. Note that blank outputs are not shorter than real ones (mean grapheme length 31.4 against 28.0), so the mechanism is repetition rather than truncation: Section 6.2 shows blank generations collapsing onto a handful of repeated strings. **The defensible claim is that real and blank CER are statistically indistinguishable**, which is sufficient for everything that follows.

Third, **the unseen-script conditions are not a distinct regime**. Ol Chiki (0.970) and Perso-Arabic (0.991) sit inside the same band as in-distribution Hindi (0.985). The model fails on real scans of its own training script as thoroughly as on scripts it has never seen.

The consequence for framing is direct. We cannot claim a dissociation between reading ability and confidence, because on real scans there is no reading ability to dissociate from. What we can claim, and what follows, is a mechanistic account of how a model in this state produces near-ceiling confidence, and a diagnostic that detects the state.

6 Confidence Under Absent and Unfamiliar Input

All four conditions sit between 0.985 and 0.990. The spread across condition means (0.0037) is smaller than the seed-to-seed spread of those means. Which seed trained the model accounts for more variation in reported confidence than whether the input is a real document, an unfamiliar script, or a blank page.

We note a statistical caution rather than leaving it to a reader. An equivalence test at a smallest-effect-of-interest of $\delta = 0.05$ is weak on this data: every condition lies within a 0.005-wide band, so equivalence within $\delta = 0.05$ is close to guaranteed by the measurement scale rather than by the

Table 3: Mean generation confidence (max-softmax of the selected token, averaged over the generated sequence). Seed columns give the mean over 60 images with the standard deviation across those images; the pooled column is over $180 = 60 \times 3$ observations on 60 shared images, not 180 independent images (Section 6.3).

Condition	Seed 0	Seed 1	Seed 2	Pooled
Hindi, real	0.982 ± 0.020	0.990 ± 0.008	0.990 ± 0.012	0.9873
Blank white	0.981 ± 0.012	0.993 ± 0.008	0.982 ± 0.015	0.9857
Ol Chiki	—	—	—	0.9857
Perso-Arabic	—	—	—	0.9894

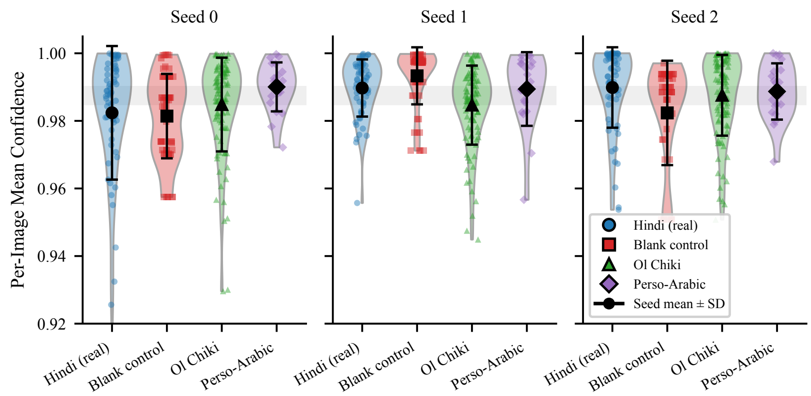

Figure 1: Per-image mean confidence by condition, faceted by seed, with the seed mean and standard deviation overlaid. The four distributions occupy the same narrow band in every seed. Seed-to-seed movement of the band is larger than any between-condition separation within a seed.

finding. The bound would be meaningful only if calibrated against the spread a grounded model exhibits over the same conditions. We therefore report the equivalence test as descriptive and rest no claim on it.

6.1 What the model emits on unfamiliar scripts

On all 360 held-out images spanning both unseen scripts and all three seeds, the model produces zero characters of the target script, substituting text composed 93–100% (per-condition mean) of Devanagari. Manual inspection shows fluent, grammatical Hindi unrelated to the source image. This is the small-scale counterpart of what GlotOCR Bench reports at scale [Kargaran et al., 2026].

6.2 Output degeneracy

Confidence equivalence across conditions is interesting only if the outputs compared are non-trivial. They are not equally so.

Table 4: Output diversity within each seed, $n = 60$ images. At seed 1 the median pairwise grapheme edit distance between outputs on *different* blank images is 0.

Condition, seed	Unique outputs / 60	Mean pairwise edit	Identical-pair rate
Hindi, seed 0	24	26.4	12%
Hindi, seed 1	30	29.4	5%
Hindi, seed 2	22	22.7	18%
Blank, seed 0	**3**	**7.5**	**34%**
Blank, seed 1	**3**	**7.2**	**76%**
Blank, seed 2	8	26.7	27%

First-token identity shows the same pattern. On blank input at seed 0 the decoder begins every one of 60 generations with the same grapheme; at seed 1, two distinct first graphemes cover all 60, one accounting for 86.7%. On real Hindi the corresponding counts are 10, 9 and 10 distinct first graphemes, with modal frequencies between 40% and 72%.

We draw the honest conclusion: part of the "confidence is unchanged on blank input" result is that the decoder emits a near-constant, highly probable string on blank input. That is a real property of the model and consistent with the paper's thesis, but it is a weaker mechanism than fluent varied hallucination and we do not describe it as the latter. Real input is also far from diverse: 22 to 30 unique outputs across 60 distinct images.

6.3 Variance structure

Because the 180 observations per condition are 60 images under three seeds, pooled standard deviations mix image variance, seed variance and their interaction. Decomposed, confidence on real Hindi is 9% image / 7% seed / 84% residual, and on blank 0% / 18% / 82%. Ground-truth likelihood behaves oppositely at 84–85% image / ≈2% seed / ≈14% residual. Which image is presented explains 9% of confidence variance on real input and none of it on blank, while explaining most of the variance in a quantity that genuinely depends on the input. This is independent evidence that the confidence signal is not responding to input content, and it means pooled standard errors in Table 3 should be read as seed-clustered.

7 Mechanism I: Ablation Moves the Distribution's Shape, Not Its Peak

We replace the encoder's output with a zero tensor of identical shape immediately before the decoder's cross-attention consumes it, and compare against the unmodified model on the same real images.

Table 5: Encoder zeroing. Image-level aggregates (left) and a step-level decomposition of the KL into positions where the argmax agrees between the full and zeroed model and positions where it flips (right). The two flip measures differ by construction: top-1 agreement is a per-image mean (12.1% disagreement pooled), while the flip rate is over all 4,990 decoding steps (8.2% pooled).

	Image-level			Step-level KL decomposition			
Seed	Δconf	KL	Top-1 agr.	Flip rate	KL agree	KL flip	Flip share
0	−0.0234	1.639	0.857	10.2%	0.050	10.62	**96.0%**
1	+0.0035	0.440	0.950	4.9%	0.017	8.92	**96.3%**
2	+0.0109	1.147	0.831	13.4%	0.010	6.48	**99.0%**
Pooled	−0.0030	1.075	0.879	8.2%	0.027	8.87	**96.7%**

We give the per-seed rows before the pooled figure deliberately. The seed-level confidence deltas span 0.034 and change sign. **The pooled −0.0030 is cancellation, not a stable small effect, and the confidence consequence of encoder ablation is unmeasured at three seeds rather than measured to be zero.**

Pooled mean confidence, split the same way: with the full encoder, 0.998 at agreeing positions and 0.893 at flipped positions; with the encoder zeroed, 0.998 and 0.825. The statement this supports, and which we make no stronger:

> At the 88% of positions where deleting the encoder does not change which token wins, the distribution is essentially unchanged (KL 0.010–0.050) and the peak stays at 0.998. At the 12% where the encoder does change the winner, the distribution changes enormously (KL 6.5–10.6) and the model remains highly confident in the new winner (0.89 with the encoder, 0.83 without). The encoder can change *which* token holds the peak. It does not change *that there is a near-unit peak.*

Since deployed confidence is the peak height, and peak height is close to invariant under the intervention that removes visual information, confidence is structurally rather than incidentally blind to this class of failure. The invariance is strong but not total: confidence at flipped positions (0.89) is meaningfully below the agreeing-position value (0.998).

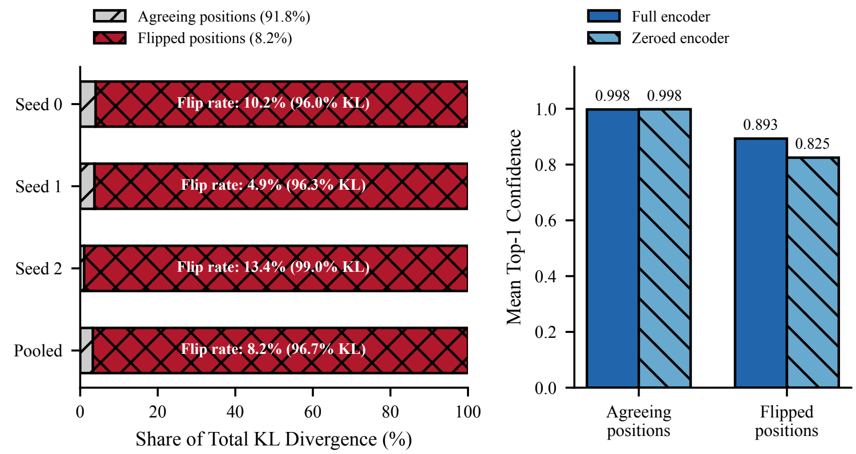

Figure 2: **Left:** share of total ablation KL contributed by agreeing versus argmax-flipping decoding steps, per seed and pooled. Flipped steps are 8.2% of the total but carry 96.7% of the divergence. **Right:** mean top-1 confidence at agreeing and flipped positions, with the full and zeroed encoder. Agreeing positions are pinned at 0.998 under both; flipped positions remain high but not pinned.

Two distinct interventions. This section zeroes the encoder output. Section 8 uses blank white images, where the encoder still runs and produces features of blankness. Zeroing asks whether the cross-attention pathway carries usable signal at all; a blank image asks whether the model distinguishes content from its absence. We keep them labelled separately.

8 Mechanism II: Teacher-Forced Likelihood and the Position Profile

Max-softmax over a model's own greedily generated tokens is upward-biased by construction. We therefore also compute teacher-forced log-likelihood of the ground-truth sequence, conditioning on true preceding tokens and recording the probability assigned to the true next token regardless of whether the model would have chosen it. Whole-sequence means are -1.783 (SD 1.274, $n = 180$) on real input and -1.751 (SD 1.280, $n = 180$) on blank, with mean predictive entropy 0.0207 and 0.0251 respectively.

The two measures disagree in sign, and we spin neither. Ground-truth likelihood is marginally higher on blank, contrary to what grounded confidence predicts. Predictive entropy is also higher on blank, meaning slightly less certain, which is exactly what grounded confidence predicts. Both differences are negligible against their dispersions: 0.032 against SDs near 1.27, and 0.004 against SDs near 0.017. Neither measure detects a difference between real and blank, and the nominal signs should not be interpreted. The two estimators are also not independent: both derive from the same softmax, differing in whether the conditioning prefix is self-generated or ground truth.

Where the ground truth wins. The ground-truth token is the model's argmax at 90.4% of positions on real input and 90.1% on blank, with a mean peak probability of 0.9983 and 0.9982 at those positions, consistent with the reported entropy. Broken out by position: **0% at position 0**, on 0 of 180 sequences, under both conditions; 92.6% (real) versus 92.3% (blank) thereafter. That split is the shape of the whole result.

Five observations, in order of importance.

1. Position 0 is far below chance, not merely uncertain. The geometric mean probability assigned to the correct first symbol is 2.2×10^{-11} on real images and 6.3×10^{-11} on blank, against a uniform value of 2.725×10^{-3}: roughly eight orders of magnitude below chance. The model is not undecided at position 0; it is near-deterministically committed to a specific wrong symbol, identically whether the image contains text or nothing. We report the geometric mean because the arithmetic mean of $p(\mathrm{GT})$ is dominated by a small number of less-severe cases and is not representative of the distribution; Appendix B gives the per-seed figures.

Table 6: Mean $\log p(\mathrm{GT})$ by generation position, with n-gram grapheme language model references fitted on 2,491 held-out-excluded training lines and evaluated from the second grapheme onward. Uniform over $|V| \approx 367$ is -5.91. Positions 2–39 track the 4-gram and 5-gram references closely; position 40+ falls back below uniform.

Positions	Real	Blank	n steps	Reference	Value
0	-24.5	-23.5	180	Uniform $(1/\|V\|)$	-5.91
1	-8.1	-8.6	180	Trigram LM	-0.99
2–9	-0.48	-0.41	1440	4-gram LM	-0.44
10–19	-0.15	-0.16	1800	5-gram LM	-0.26
20–39	-0.27	-0.28	2637		
40+	-6.12	-6.28	1134		

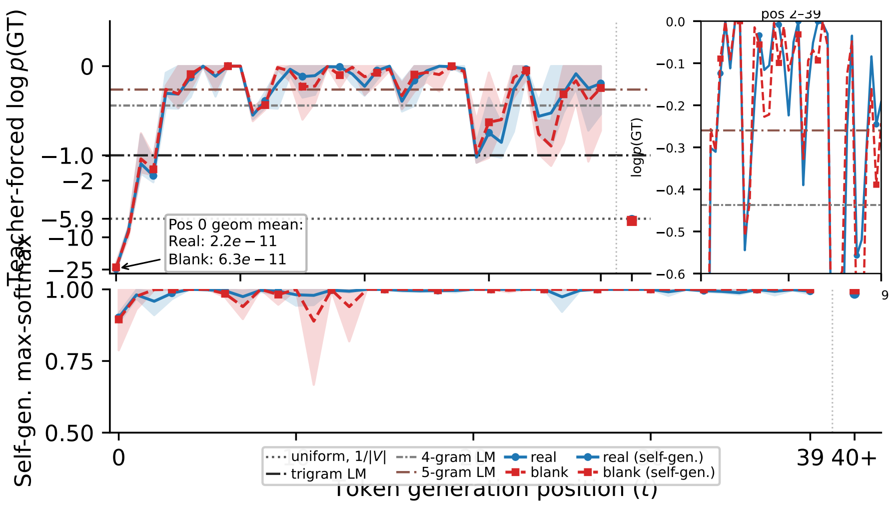

Figure 3: The dissociation. **Top:** mean $\log p(\mathrm{GT})$ against generation position, real and blank overlaid, with horizontal references at uniform (-5.91), trigram (-0.99), 4-gram (-0.44) and 5-gram (-0.26) grapheme language models. The inset (right-hand panel) zooms positions 2–39 to separate the 4-gram and 5-gram references, and the rightmost point aggregates positions 40 and beyond over 1,134 steps per condition. **Bottom:** self-generated max-softmax confidence at the same positions. Correctness collapses at positions 0–1 while confidence does not, and both curves are near-identical between real and blank input.

2. Confidence at that position is near ceiling. Self-generated max-softmax at position 0 is 0.902 pooled on real input and 0.895 on blank. Position 0 is therefore the point of maximal confidence and minimal correctness, at the one generation step where vision is the only possible information source. Per seed, real: 0.884 / 0.906 / 0.916; blank: 0.9999 / 0.898 / 0.787. The blank column is not stable across seeds and its pooled value should not be quoted alone.

3. Position 1 is also below chance. At -8.1 and -8.6, the second generated symbol remains more than two nats below the uniform reference. The failure is not confined to a single step; recovery begins at position 2.

4. In the middle of the sequence the model behaves as a high-order grapheme n-gram model, and nothing more. Fitting n-gram grapheme language models on the training text with the evaluation strings excluded gives rest-of-sequence means of -4.17 (unigram), -2.25 (bigram), -0.99 (trigram), -0.44 (4-gram) and -0.26 (5-gram). The instrument's positions 2–9 (-0.48) sit essentially on the 4-gram reference, and positions 20–39 (-0.27) essentially on the 5-gram reference. We therefore do *not* claim the decoder has learned something beyond local n-gram statistics; the honest reading is that its mid-sequence behaviour is well approximated by a 4- to 5-gram grapheme prior, achieved without reference to the image, since the blank column matches the real column at every bucket.

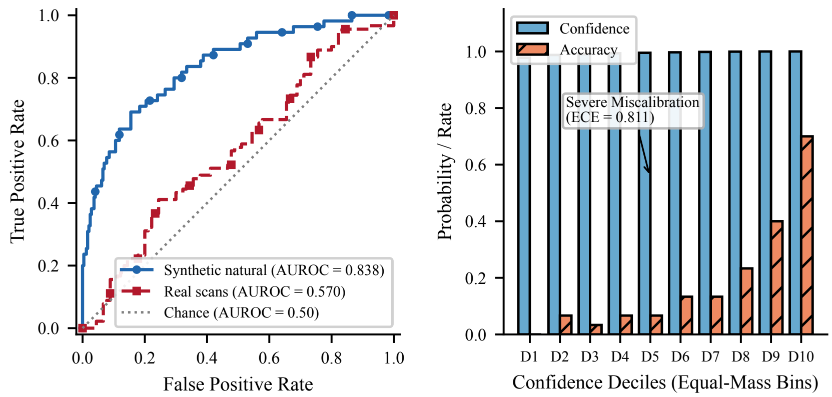

Figure 4: **Left:** confidence as a ranker of correctness in two regimes. On synthetic in-distribution lines it separates correct from incorrect transcriptions; on real scans it is close to chance. **Right:** reliability diagram on synthetic natural lines, equal-mass confidence deciles. Confidence is near unity in every decile while accuracy is not, so the model discriminates without being calibrated.

5. A second collapse past position 40. Beyond position 39 the mean falls to -6.12 (real) and -6.28 (blank), below the uniform reference, on 1,134 steps per condition. Long sequences are those on which the model has drifted furthest from any prefix the training distribution supports, and there the n-gram prior stops helping. This bucket is reported for completeness; we place no weight on it, since sequences reaching position 40 are a length-selected minority.

9 When the Confidence Signal Does Work

A purely negative result would leave open the possibility that this model's confidence head is simply broken. It is not, and the contrast is informative.

Table 7: Confidence as a predictor of correctness across evaluation regimes. AUROC is against line-level exact correctness except in the final row, which uses a median split on grapheme CER because exact accuracy is identically zero.

Evaluation set	Line acc.	AUROC	Spearman (conf, CER)
Synthetic, natural exposure	0.183	**0.838**	-0.40
Synthetic, flattened / inverted	0.003 / 0.007	undefined	≈ 0
Real Hindi scans	0.000	undefined	-0.15
Real Hindi scans, median CER split	—	0.57	—

On synthetic in-distribution lines, where the model reads well enough to get 18.3% of lines exactly right, confidence ranks correct transcriptions above incorrect ones at AUROC 0.838. That is a working discrimination signal (per seed 0.763, 0.924, 0.832), and it is nonetheless badly calibrated: mean confidence in that regime is 0.994 against 18.3% accuracy, giving an expected calibration error of 0.812 over ten equal-mass bins. Discrimination and calibration come apart cleanly here, and only the former survives. On real scans, where accuracy is zero, a graded substitute gives 0.57, near chance. Mean confidence across the three synthetic exposure conditions also varies substantially, at 0.994 (natural), 0.600 (flattened) and 0.461 (inverted), tracking the accuracy drop in those conditions.

Taking this together with Sections 5–8, the reading best supported by the evidence is that **the model's confidence responds to properties of the text it is generating, not to whether the image supports that text.** It falls when the target text has unfamiliar statistics. It does not fall when the image is blank, unreadable, or in a script the model has never seen, because in those cases the decoder still produces fluent, familiar Devanagari. We offer this as an interpretation consistent with what we measured, not as a demonstrated mechanism.

10 Limitations and Future Work

No positive control; this is the central limitation. We have no model with demonstrated Devanagari reading ability on which the position-0 diagnostic returns a different answer. Without that contrast the diagnostic has no measured dynamic range, and a reader cannot distinguish a working instrument from a broken one on this evidence alone. The natural control is a model with independently verified real-scan Devanagari competence [Singh, 2026], on which position-resolved teacher-forced $\log p(\mathrm{GT})$ is directly computable from open weights. We regard this as the next experiment rather than an optional extension.

Three further interventions we consider necessary. First, *mismatched pairing*: teacher-forced $\log p(\mathrm{GT})$ under a derangement of the image pool, so each ground-truth string is scored against a different real image. A blank page is a degenerate input, so "blank equals real" is weaker evidence than "wrong real image equals right real image" would be, and the latter would remove the degeneracy confound of Section 6.2 entirely. Second, *cross-attention contribution norms*, $\|\text{cross-attn out}\|/\|\text{residual}\|$ per decoder layer, which would convert the behavioural claim that the model ignores the encoder into a direct measurement of how much signal that pathway carries. Third, *Gaussian-noise and patch-scrambled inputs*, testing whether the model responds to visual variation of any kind rather than only to the blank versus real contrast.

Scale and generality. One model of approximately 19.5M parameters, one training script for the mechanistic experiments. We make no claim about production OCR systems. A separate attempt to use TrOCR-base-printed as an external comparison found mean confidence 0.42 on its own in-domain English text, far below this instrument's near-ceiling regime; that is an operating-regime mismatch rather than a failed replication, and it does not serve as a control.

Metric choice. Line-level exact match is uninformative here and is used only in Section 9, where the comparison is between regimes rather than between conditions. Grapheme CER is what Section 5 reports.

The exposure-versus-complexity question is unanswered. We believe the fix is to abandon distributional-target synthesis in favour of subsampling real, naturally learnable sentences to bias exposure, accepting weaker distributional control in exchange for text that remains learnable.

Novelty relative to prior work. That a multimodal model's confidence is insensitive to visual ablation is established [Khanmohammadi et al., 2026]. Our contribution is the positional localisation of that insensitivity in sequence transcription, the shape-versus-peak decomposition of the ablation, and the within-instrument regime contrast in Section 9, not the insensitivity itself.

11 Discussion and Conclusion

Three implications follow from the position profile, if it holds under a positive control. Visual dependence in autoregressive transcription is concentrated at the start of the sequence: positions 2 onward are recoverable from the prefix alone at 62–86% per-token likelihood, above a trigram baseline, and identically so with no image at all. Any confidence score averaged uniformly over the generated sequence therefore averages the one vision-dependent position against many that are not, diluting exactly the signal a gating system needs; confidence measures for transcription should weight early positions. The diagnosis is also cheap: two forward passes and ground-truth text give the position profile, with no probe training, no auxiliary labels and no comparison model. Where probe-based approaches [Khanmohammadi et al., 2026, Yao et al., 2025, Kogilathota et al., 2026] detect ungroundedness by learning it, the position-0 test asks a structural question with a known answer under grounding. And the failure mode is a training pathology that standard monitoring would not catch: a model in this state converges, produces fluent in-script output, and reports mean confidence of 0.99.

We audited a small, fully exposure-controlled document OCR model and found that it does not read real document images. We localised the failure to the start of the generated sequence, where the ground-truth symbol is never the model's argmax and receives roughly eight orders of magnitude less probability than chance, identically with and without visual content, while self-reported confidence at that position is approximately 0.90. We showed that deleting the encoder alters the output distribution's shape substantially while leaving its peak near unity, so that the quantity read as confidence is close to invariant under precisely the intervention that removes visual information. And we showed that the same confidence signal does discriminate correctness in a regime where the model reads, locating the failure in the relationship between confidence and vision rather than in the confidence head itself.

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A A Retracted Single-Seed Result

An earlier analysis using only seed 0 reported a statistically significant confidence difference on the Perso-Arabic condition relative to in-distribution Hindi ($|z| = 2.54$). It did not replicate: at seed 1, Hindi confidence (0.9897) exceeded Perso-Arabic (0.9894), reversing the original direction. We retract the claim in full and report it as the failure mode that motivated the three-seed design.

B Position-0 Likelihood Distribution

Per-seed geometric means of $p(\mathrm{GT})$ at position 0 are 2.05×10^{-10} / 1.35×10^{-11} / 3.88×10^{-12} (real) and 9.32×10^{-10} / 3.13×10^{-11} / 8.70×10^{-12} (blank). The effect is present independently at every seed and spans two orders of magnitude across seeds, so per-seed values rather than the pooled geometric mean should be used when comparing against another model.

A geometric mean can be driven by a minority of extreme cases, so we report the distribution. Pooled over 180 sequences, the median $p(\mathrm{GT})$ at position 0 is 9.5×10^{-15} (real) and 3.7×10^{-13} (blank); the arithmetic means are 2.67×10^{-3} and 1.51×10^{-3}, close to uniform, but they are set by a handful of sequences. Only 2.8% of real sequences (5 of 180) and 3.3% of blank sequences (6 of 180) assign the correct first symbol more probability than uniform.

Table 8: Histogram of $\log p(\mathrm{GT})$ at position 0, pooled over three seeds.

Bin ($\log p$)	Real count	Real %	Blank count	Blank %
< -25	126	70.0%	115	63.9%
$[-25, -20)$	24	13.3%	25	13.9%
$[-20, -15)$	14	7.8%	13	7.2%
$[-15, -10)$	8	4.4%	13	7.2%
$[-10, -5)$	6	3.3%	10	5.6%
$[-5, 0]$	2	1.1%	4	2.2%

Seventy percent of real sequences place $\log p(\mathrm{GT}) < -25$ at position 0. The collapse is a property of the distribution, not of its summary.

C Paired Tests and Bootstrap Intervals

Because the same 60 image identifiers appear under every condition, we use paired rather than unpaired tests. Per-seed Wilcoxon signed-rank on grapheme CER (real minus blank) gives $+0.180$ ($p = 1.4 \times 10^{-6}$), $+0.088$ ($p = 0.106$) and -0.159 ($p = 0.015$) at seeds 0, 1 and 2. The sign reverses at seed 2. Pooled unclustered the difference is $+0.036$ ($p = 0.025$); clustered by seed it is not significant ($t = 0.257$, $p = 0.822$). For mean confidence the pooled difference is $+0.0016$ ($t = 0.505$, $p = 0.664$).

Seed-clustered bootstrap intervals (2,000 replicates, images resampled within seed): grapheme CER 0.9855 [0.9406, 1.0347] real and 0.9491 [0.9053, 0.9911] blank; mean confidence 0.9873 [0.9852, 0.9892] real, 0.9857 [0.9838, 0.9874] blank, 0.9857 [0.9843, 0.9871] Ol Chiki and 0.9894 [0.9871, 0.9916] Perso-Arabic; position-0 geometric mean $p(\mathrm{GT})$ 2.20×10^{-11} [1.04×10^{-11}, 5.29×10^{-11}] real; AUROC 0.838 [0.771, 0.902] on synthetic natural and 0.570 [0.491, 0.650] on real scans.

D Sequence Lengths

Mean and median generated grapheme lengths are 27.96 / 28.0 (real Hindi), 31.39 / 29.0 (blank), 28.28 / 29.0 (Ol Chiki) and 33.40 / 35.0 (Perso-Arabic), n as in Table 2. Blank generations are not shorter than real ones, which rules out truncation as the explanation for the CER ordering in Section 5 and leaves repetition, documented in Section 6.2, as the operative mechanism.